Supplementary Methods: simulations

Supplementary code documentation for:

Multistability in species-rich communities

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`https://github.com/GuimAguade/MultistabilitySynthesis`

This document describes the models, parameters, initial conditions, perturbation protocols, numerical procedures and stability criteria implemented in the Python scripts used to produce the simulation panels in Figure 2 and the three-species HOI model discussed in the main text. Each script is self-contained and can be run independently. The codes are organized by figure panel. Figures 1 and 3 are conceptual diagrams produced without simulation code. All scripts require `numpy`, `scipy`, and `matplotlib`; `fig_HOI_C.py` additionally uses `scipy.optimize`.

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1. General protocol

Pulse perturbation design

All time-series panels (Fig. 2A–F and Fig. 1B) follow the same design. (i) The system is integrated from a specified initial condition over a pre-perturbation window. (ii) At the perturbation time t_p (dashed vertical line in each panel), the state vector is changed instantaneously while all parameters are kept fixed (a pulse perturbation). (iii) Integration is restarted from the perturbed state over a post-perturbation window, and the two phases are concatenated into a single time series. A perturbation is *additive* when a fixed amount Δ is added to the abundance of the targeted species, $N_i(t_p^+) = \max\big(N_i(t_p^-) + \Delta, 0\big)$, and *multiplicative* when their abundance is rescaled, $N_i(t_p^+) = f\, N_i(t_p^-)$. Clipping at zero is applied in all additive perturbations except in Fig. 2C.

Stability criteria

An equilibrium $\mathbf{N}^*$ is called feasible if $N_i^* > 0$ for all species considered present, and locally (linearly) stable if all eigenvalues of the Jacobian $J_{ij} = \partial \dot{N}_i / \partial N_j$ evaluated at $\mathbf{N}^*$ have negative real parts. For boundary equilibria (some species absent), the Jacobian includes the rows of the absent species, whose diagonal entries are their invasion growth rates; a boundary equilibrium is therefore stable only if it is both internally stable and non-invadable by the absent species. A system is called multistable when at least two distinct equilibria are simultaneously feasible and stable. In the time-series panels, the states reached at the end of each integration phase are interpreted as attractors of the dynamics; for the models with closed-form equilibria (2A, 2C, 2D, HOI) the corresponding analytical stability conditions are given below. In the high-dimensional GLV panel (2F), stable states are identified dynamically by convergence from random initial conditions (Section 7). In the HOI phase diagram, stability is assessed explicitly from Jacobian eigenvalues (Section 8).

Numerical integration

All ODEs are integrated with `scipy.integrate.solve_ivp` using the explicit Runge–Kutta 4(5) method (RK45). Unless otherwise stated, SciPy's default tolerances are used (relative tolerance 10^{-3}, absolute tolerance 10^{-6}). Panels 2B, 2D and 2E use tightened tolerances (rtol $= 10^{-8}$, atol $= 10^{-10}$). Solutions are stored on evenly spaced output grids (`t_eval`), which set the plotted resolution but not the integration step.

2. Figure 2A — Single-species Allee effect (`fig2A.py`)

Model

The single-species (strong) Allee effect is described by:

$$\frac{dN}{dt} = f(N) = rN\left(1 - \frac{N}{K}\right)\left(\frac{N}{A} - 1\right) \tag{1}$$

where N is population abundance, r the intrinsic growth rate, K the carrying capacity, and A the Allee threshold. Parameters: $r = 1.0$, $K = 1.0$, $A = 0.5$.

Equilibria and stability

The model has three equilibria, $N^* \in \{0, A, K\}$, with linearizations $f'(0) = -r$, $f'(A) = r(1 - A/K)$ and $f'(K) = -r(K/A - 1)$. For $0 < A < K$, $N^* = 0$ and $N^* = K$ are stable and $N^* = A$ is unstable, so that

$N = A$ separates the basins of attraction of the two stable states.

Simulation and perturbation protocol

A single trajectory is initialized at $N_0 = 0.65$ and integrated over $t \in [0, 10]$ with default RK45 tolerances and 500 output points. At $t_p = 10$, an additive pulse $\Delta N = -0.52$ is applied (clipped at zero), and integration continues over $t \in [10, 25]$ with 500 output points.

The script also contains (after the `exit()` call) code to generate a 1D phase portrait, used for reference during figure preparation. Flow direction is given by the sign of $f(N)$ at 20 evenly spaced points in $[0.05, 1.15]$, and each equilibrium N^* is classified as stable if $f(N^* + 10^{-4}) < 0$ and unstable otherwise.

3. Figure 2B — Two-species plant–pollinator model (`fig2B.py`)

Model

A minimal mutualistic system of one plant (P) and one pollinator (A), inspired by a two-dimensional simplification of the model framework of Lever et al. (2014):

$$\frac{dP}{dt} = rP + P \cdot \frac{\gamma A}{1 + h\gamma A} - CP^2 + \mu \tag{2}$$

$$\frac{dA}{dt} = (r - d)A + A \cdot \frac{\gamma P}{1 + h\gamma P} - CA^2 + \mu \tag{3}$$

Mutualistic benefits saturate via a Holling type II functional response with handling time h. The parameter d introduces an asymmetry in intrinsic growth, representing net mortality of the pollinator in the absence of the plant ($r - d < 0$ for the parameters used). Intraspecific competition is quadratic with strength C, and a small immigration term μ prevents exact extinction. Parameters: $r = 0.2$, $d = 0.97$, $\gamma = 3.0$, $h = 0.5$, $C = 1.0$, $\mu = 10^{-4}$.

Simulation and perturbation protocol

Both species are initialized at $P_0 = A_0 = 0.5$ and integrated over $t \in [0, 50]$ (rtol $= 10^{-8}$, atol $= 10^{-10}$, 1000 output points). At $t_p = 50$, an additive pulse $\Delta A = -0.35$ is applied to the pollinator only (clipped at zero), the plant abundance being left unchanged. Integration continues over $t \in [50, 100]$ with the same solver settings.

4. Figure 2C — Two-species competitive Lotka–Volterra (`fig2C.py`)

Model

Classical competitive Lotka–Volterra system, as presented in Petraitis (2013):

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1}\right) \tag{4}$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2}\right) \tag{5}$$

Parameters: $r_1 = r_2 = 1.0$, $K_1 = K_2 = 1.0$, $\alpha_{12} = 1.1$, $\alpha_{21} = 1.2$.

Equilibria and stability

The system has four equilibria. The trivial state $(0, 0)$ has eigenvalues $r_1, r_2 > 0$ and is unstable. The single-species state $(K_1, 0)$ has eigenvalues $-r_1$ and $r_2(1 - \alpha_{21} K_1/K_2)$, the latter being the invasion growth rate of species 2; it is stable when $\alpha_{21} > K_2/K_1$. Symmetrically, $(0, K_2)$ is stable when $\alpha_{12} > K_1/K_2$. The interior equilibrium is

$$N_1^* = \frac{K_1 - \alpha_{12} K_2}{1 - \alpha_{12}\alpha_{21}}, \qquad N_2^* = \frac{K_2 - \alpha_{21} K_1}{1 - \alpha_{12}\alpha_{21}}, \tag{6}$$

which, for $K_1 = K_2$ and $\alpha_{12}, \alpha_{21} > 1$, is feasible and a saddle. In this regime both single-species states are stable, and the stable manifold of the saddle separates their basins of attraction.

Simulation and perturbation protocol

The system is initialized at $(N_1, N_2) = (0.4, 0.6)$ and integrated over $t \in [0, 75]$ with default RK45 tolerances and 500 output points. At $t_p = 75$, the species with the higher abundance is identified, and an additive pulse of -0.8 is applied to it (without clipping). Integration continues over $t \in [75, 100]$ with 500 output points.

Phase plane

The script also generates a phase plane of the same system. The vector field $(\dot{N}_1, \dot{N}_2)$ is evaluated on a 14×14 grid over $[0, 1.2]^2$ and normalized to unit length, so that only flow direction is represented, and is displayed as a quiver plot.

5. Figure 2D — Multi-species coupled Allee effect (`fig2D.py`)

Model

An S-species generalization of the single-species Allee model, where species are coupled through a facilitation matrix $\mathbf{C}$:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i}{K_i}\right)\left(\frac{w_i}{A_i} - 1\right), \qquad w_i = \sum_j C_{ij} N_j \tag{7}$$

The Allee term depends on a weighted community abundance w_i, capturing the idea that each species benefits from the collective presence of others. With $C_{ii} = 1$ and $C_{ij} = \epsilon$ for $i \neq j$, the coupling is weak but positive and links the bistable dynamics of all species. Because the growth rate of species i is proportional to N_i, an extinct species cannot be rescued by facilitation from the others.

Parameters

Five species are simulated ($S = 5$), with $\epsilon = 0.1$. Parameters r_i, K_i, A_i are drawn independently from normal distributions with standard deviation 0.1 and means matching the 1D model ($\mu_r = 1.0$, $\mu_K = 1.0$, $\mu_A = 0.5$). The random seed is set with `numpy.random.seed(42)`, and $\mathbf{r}$, $\mathbf{K}$, $\mathbf{A}$ are drawn in that order.

Equilibria and stability

At $\mathbf{N}^* = \mathbf{K}$, the factor $(1 - N_i/K_i)$ vanishes for all i, so the Jacobian is diagonal with entries $J_{ii} = -r_i\,(w_i/A_i - 1)$, where $w_i = K_i + \epsilon \sum_{j \neq i} K_j$; this state is stable if $w_i > A_i$ for all i. The extinction state $\mathbf{N}^* = \mathbf{0}$ has a diagonal Jacobian with entries $-r_i$ and is always stable. Mixed states, with a subset of species at K_i and the remaining species at zero, are also equilibria, and their stability follows from the same diagonal structure.

Simulation and perturbation protocol

All species are initialized at $N_i = 0.65$ and integrated over $t \in [0, 10]$ (rtol $= 10^{-8}$, atol $= 10^{-10}$, 500 output points). At $t_p = 10$, an additive pulse of $-4/S = -0.8$ is applied to every species simultaneously (clipped at zero), and integration continues over $t \in [10, 25]$ with the same solver settings. Within the right-hand side, negative abundances are set to zero before evaluation.

6. Figure 2E — Multi-species plant–pollinator community (`fig2E.py`)

Model

Extension of the two-species mutualistic model (Section 3) to n_P plant and n_A pollinator species:

$$\frac{dP_i}{dt} = r_{P,i} P_i + P_i \cdot \frac{\sum_j \gamma_{ij} A_j}{1 + h \sum_j \gamma_{ij} A_j} - C P_i^2 + \mu \tag{8}$$

$$\frac{dA_j}{dt} = (r_{A,j} - d) A_j + A_j \cdot \frac{\sum_i \gamma_{ij} P_i}{1 + h \sum_i \gamma_{ij} P_i} - C A_j^2 + \mu \tag{9}$$

Parameters

The interaction matrix $\boldsymbol{\gamma}$ is fully connected and deterministic: all plant–pollinator pairs interact with strength $\gamma_{ij} = 1.2$, except the pairs $i = j$, for which $\gamma_{ii} = 0.5$. It is therefore not nested and is kept simple for illustration. Species-specific growth rates $r_{P,i}$ and $r_{A,j}$ are drawn from Uniform$(0.15, 0.25)$ using `numpy.random.default_rng(seed=42)`, plants first and pollinators second. Other parameters: $n_P = n_A = 4$, $d = 1.215$, $h = 0.5$, $C = 1.0$, $\mu = 10^{-4}$.

Simulation and perturbation protocol

Initial abundances of all eight species are drawn from Uniform$(0.4, 0.6)$ using `numpy.random.uniform`. The system is integrated over $t \in [0, 30]$ (rtol $= 10^{-8}$, atol $= 10^{-10}$, 1000 output points). At $t_p = 30$, an additive pulse $\Delta A_j = -0.3$ is applied to all pollinators simultaneously (clipped at zero), plant abundances being left unchanged. Integration continues over $t \in [30, 100]$ with the same solver settings.

7. Figure 2F — Multi-species generalized Lotka–Volterra (`fig2F.py`)

Model

The generalized Lotka–Volterra (GLV) model with S species, as in Aguadé-Gorgorió & Kéfi (2024):

$$\frac{dx_i}{dt} = x_i\left(1 - x_i + \sum_j A_{ij}x_j\right) \tag{10}$$

Interaction coefficients A_{ij} $(i \neq j)$ are drawn independently from $\mathcal{N}(\mu_A, \sigma_A)$, with $A_{ii} = 0$ so that self-regulation is contained in the term $-x_i$. For mean competition ($\mu_A < 0$) and sufficiently heterogeneous interactions, this model can admit multiple alternative stable states, i.e. distinct subsets of coexisting species. Parameters: $S = 50$, $\mu_A = -0.75$, $\sigma_A = 0.5$.

Numerical procedure and extinction threshold

Integration uses RK45 with default tolerances, a maximum step size of 1 time unit and dense output, evaluated on a grid of two points per time unit. To handle extinctions, the time derivative of any species with $x_i \leq \varepsilon = 10^{-8}$ is replaced by $\max(0, \dot{x}_i)$. Species below ε therefore cannot decline further but can regrow if their growth rate is positive, so that ε acts as an abundance floor rather than an absorbing extinction.

Identification of alternative stable states

Alternative stable states are identified dynamically on a single interaction matrix $\mathbf{A}$. In each trial, initial abundances for all S species are drawn from Uniform$(0, 1)$ and the system is integrated over $t \in [0, 2000]$. The state at $t \approx 1998$ (the fifth-to-last output point) is used as initial condition for a second integration of 10 time units, and the trial is accepted as converged if the corresponding states of the two integrations agree within an absolute tolerance of 10^{-4} (`numpy.allclose`, relative tolerance 10^{-5}). Survivors are defined as species with abundance $> 10^{-4}$ in the converged state, and trials with no survivors are discarded. Because absent species are held at the floor ε and can regrow during the long integration, the convergence criterion selects states that are both dynamically stable and non-invadable by the rest of the pool, up to numerical precision; no Jacobian is evaluated. Trials are repeated until a converged state is found whose survivor set differs from that of the first converged state, up to a maximum of 500 trials. Two states are considered distinct if their survivor sets are not identical.

Competition and perturbation experiments

The two survivor sets (states 1 and 2) are merged into a combined system consisting of the union of their species, of size $N \leq S$, with interaction matrix given by the corresponding submatrix of $\mathbf{A}$; pool species absent from both states are excluded. Two complementary experiments are performed. In experiment 1, species of state 1 are initialized with abundances drawn from Uniform$(0.8, 1.2)$ and species of state 2 with abundances drawn from Uniform$(0.1, 0.3)$; species belonging to both states are assigned the latter value. The system is integrated over $T_{\text{pre}} = 40$ time units. At $t_p = T_{\text{pre}}$, a multiplicative pulse with $f = 0.2$ is applied to all species of state 1 (including shared species), and post-perturbation dynamics are integrated over $T_{\text{post}} = 20$ time units. Experiment 2 is identical with the roles of states 1 and 2 exchanged. The random seed (`numpy.random.seed(42)`) is set after the identification of the alternative states and before the generation of the initial conditions of the two experiments.

8. Three-species HOI model — time series and phase diagram (`fig_HOI_B.py`, `fig_HOI_C.py`)

Overview

Nonlinear dynamics are a well-known mechanism by which a simple system can be multistable. In GLV models, interactions of the form $N_i A_{ij} N_j$ are linear in the abundance of each partner. Here we study a 3-species toy model with logistic growth and a third-order interaction, in the limit-case assumption where species interact only when the full triplet is present, as in consortia in which the interactions perceived by a species are mediated by two other species at the same time (Fig. 1A):

$$\frac{dN_i}{dt} = r_i N_i\left(1 - \frac{N_i - B_{ijk}N_jN_k}{K_i}\right) \tag{11}$$

where the effect on species i depends on the *product* of the abundances of the other two species j and k. This introduces a nonlinearity absent from pairwise Lotka–Volterra models. In all simulations, $r_i = K_i = 1$, and we write $B_i \equiv B_{ijk}$ for the single HOI coefficient acting on species i, with $B_i < 0$ corresponding to competition. The model is used to test whether bistability between a three-species state and a two-species state can arise at the scale of the triplet, and is compared against a pairwise GLV model with the same number of species.

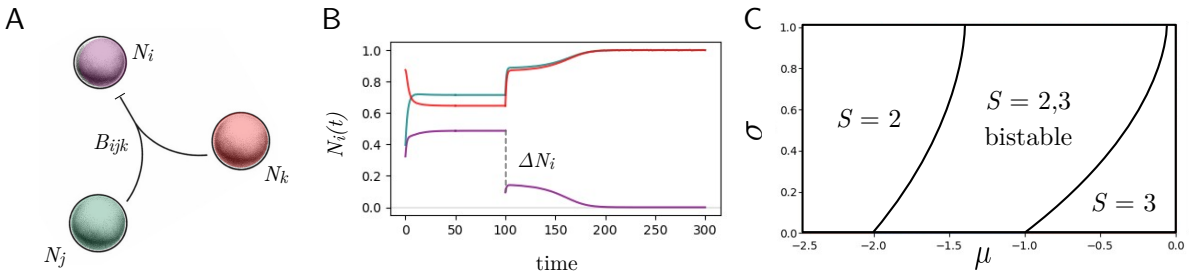


Figure 1: (A) Diagram for a three-species higher-order interaction. (B) Time series of the HOI model with a pulse perturbation on species 1 (parameters: $B_1 = -1.112$, $B_2 = -0.908$, $B_3 = -1.018$; $r_i = K_i = 1\ \forall i$). (C) Monte Carlo estimate of the probability of multistability between an $S = 3$ and an $S = 2$ state as a function of the mean μ and standard deviation σ of the HOI coefficients (see `https://github.com/GuimAguade/MultistabilitySynthesis` for the code).

Equilibria and stability

Boundary equilibria. If species i is absent, the HOI terms of the other two species vanish and each of them follows logistic growth with equilibrium $N = 1$. The boundary equilibrium with species i absent (e.g. $(0, 1, 1)$ for $i = 1$) is therefore feasible for any $\mathbf{B}$. Its Jacobian has eigenvalues -1, -1 and $1 + B_i$, the last being the invasion growth rate of species i, so it is stable if and only if $B_i < -1$.

Interior equilibria. Interior equilibria satisfy $N_i = 1 + B_i N_j N_k$. Eliminating N_1 and N_2,

$$N_2 = \frac{1 + B_2 N_3}{1 - B_1 B_2 N_3^2}, \qquad N_1 = 1 + B_1 N_2 N_3, \tag{12}$$

and substituting into the equation for N_3 yields the quintic

$$B_1^2 B_2^2 N_3^5 - B_1^2 B_2^2 N_3^4 - 2B_1 B_2 N_3^3 + B_1 B_2 (2 - B_3) N_3^2 + (1 - B_3(B_1 + B_2))\, N_3 - (1 + B_3) = 0. \tag{13}$$

Real positive roots N_3 are retained (singular cases with $|1 - B_1 B_2 N_3^2| < 10^{-12}$ are skipped), and N_2 and N_1 are reconstructed from the expressions above. Feasibility requires all three abundances to be positive. Stability is assessed from the eigenvalues of the Jacobian, with entries $J_{ii} = 1 - 2N_i + B_i N_j N_k$ and $J_{ij} = B_i N_i N_k$ $(k \neq i, j)$.

Simulation — time series (`fig_HOI_B.py`)

The HOI coefficients are set to $B_1 = -1.112$, $B_2 = -0.908$, $B_3 = -1.018$. The script first computes the candidate interior equilibria from the roots of the quintic (`numpy.roots`) and requires at least two of them to proceed; these equilibria are computed for reference and are not used to initialize the simulation. Initial abundances are drawn from $\mathcal{N}(0.4, 0.2)$. To enforce extinction, abundances $N_i \leq \varepsilon = 10^{-10}$ are set to zero together with their time derivative, so that extinction is absorbing. Integration uses RK45 with default tolerances and 2000 output points per phase, in three phases: an initial transient over $t \in [0, 50]$; a continuation from the final state, without perturbation, over $t \in [50, 100]$; and a perturbation phase over $t \in [100, 300]$, started by setting the abundance of species 1 to $N_1 = 0.095$ while species 2 and 3 are left unchanged.

Simulation — phase diagram (`fig_HOI_C.py`)

For each combination of mean interaction strength μ and standard deviation σ, the script estimates by Monte Carlo sampling, for both the HOI model and a pairwise GLV model, the probability that the system admits: (i) a feasible and stable interior equilibrium (all three species present), (ii) a feasible and stable boundary equilibrium with species 1 absent, and (iii) both simultaneously, which defines multistability. A 300×300 grid is scanned over $\mu \in [-3, 0.5]$ and $\sigma \in [0, 1.5]$, and at each grid point $n = 500$ independent random interaction structures are sampled for each model. No random seed is set.

GLV. The pairwise model is $dN_i/dt = N_i(1 - N_i + \sum_j A_{ij} N_j)$ with $S = 3$ and $A_{ii} = 0$. Off-diagonal entries A_{ij} are drawn from $\mathcal{N}(\mu, \sigma)$, and the matrix is symmetrized by setting $A_{ij} = A_{ji}$ for $i < j$. The interior equilibrium is obtained by solving $(\mathbf{I} - \mathbf{A})\mathbf{N} = \mathbf{1}$, and the boundary equilibrium by solving the corresponding 2×2 system for species 2 and 3; samples with singular matrices are skipped. Stability is assessed from the eigenvalues of the full 3×3 Jacobian, with entries $J_{ii} = 1 - 2N_i + \sum_j A_{ij} N_j$ and $J_{ij} = N_i A_{ij}$.

HOI. Coefficients B_i are drawn independently from $\mathcal{N}(\mu, \sigma)$. The boundary equilibrium is evaluated at $(N_1, N_2, N_3) = (0, 1, 1)$, which is exact for any $\mathbf{B}$, and its stability is assessed from the Jacobian eigenvalues.

The interior equilibrium is found numerically with `scipy.optimize.root` (default `hybr` method) from the initial guess $(1, 1, 1)$; it is retained if the solver reports convergence and all abundances are positive, and its stability is assessed from the Jacobian eigenvalues. A single root is sought per sample.

Stored quantities. For each grid point, feasibility and multistability are stored as fractions of all n samples, and stability as fractions of the feasible samples. The multistability matrices of the two models are displayed as heatmaps over the (μ, σ) plane.

Dependencies

All scripts were developed and tested with Python 3.10. Required packages: `numpy`, `scipy`, `matplotlib`. No installation beyond standard scientific Python is required. Scripts can be run directly from the command line (`python fig2A.py`) or from any interactive environment.

Repository structure

File	Figure
`fig2A.py`	Figure 2A: single-species Allee effect
`fig2B.py`	Figure 2B: two-species plant–pollinator model
`fig2C.py`	Figure 2C: two-species competitive Lotka–Volterra
`fig2D.py`	Figure 2D: multi-species coupled Allee effect
`fig2E.py`	Figure 2E: multi-species plant–pollinator community
`fig2F.py`	Figure 2F: multi-species generalized Lotka–Volterra
`fig_HOI_B.py`	Three-species HOI model: time series
`fig_HOI_C.py`	Three-species HOI model: phase diagram